Dropbox's Magic Pocket: leaving Amazon S3 for an in-house blob store

IT457 Cloud Computing · Case study, Week 1 · 202611054, 202611032, 202611009

21 October 2026

Between 2013 and 2016 Dropbox moved more than 90% of roughly 600 PB of user files off Amazon S3 into Magic Pocket, a storage system it designed, wrote and runs in leased co-location space. Below are the points that matter for the case, followed by the sources we found.

Why leave S3

From its first year Dropbox kept metadata and web servers in its own data centres and only the file contents on S3.[1] By 2015 those contents had grown to about 500 PB, twelve times the 40 PB of 2012.[1] The workload was narrow: immutable blocks of up to 4 MB, written once and mostly read while fresh. Over 40% of reads touch data from the last day and over 90% data from the last year.[4] A general-purpose object store charges for flexibility that such a workload never uses, and because storage is most of Dropbox's cost of revenue, the S3 bill effectively set the company's gross margin. At this size, the fixed cost of a dedicated team and custom hardware could be paid back, and owning the stack also meant control over disk format, erasure code and repair policy.

How the move was run

After a dark launch in August 2014, every write went to both systems and spare copies stayed on S3 for six months. The bulk copy, called BASE Jump, peaked above half a terabit per second, and the 90% mark was reached on 7 October 2015, three weeks early and with no data lost.[1] The duplicated data was not free: the S-1 notes higher storage cost until the last S3 copies were retired in Q4 2016.[8] Part way through, the disk-facing storage engine was rewritten from Go in Rust, because Go's memory footprint was too large for dense storage servers.[11]

Summer 2013	project starts
Aug 2014	dark launch, dual writes
27 Feb 2015	serves users
30 Apr 2015	BASE Jump bulk copy
7 Oct 2015	90% in-house
Q4 2016	S3 copies retired

Table 1: Migration timeline.[1,8]

Architecture

Files are split into blocks of at most 4 MB, each named by its SHA-256 hash. Blocks are packed into 1 GB *buckets*, and buckets into *volumes* that sit on a set of storage servers (OSDs). Two MySQL tables hold all placement state: the sharded Block Index maps a hash to its cell and bucket, and the Replication Table maps a bucket to its volume and disks.[2] The OSDs themselves are kept simple. A Master per cell

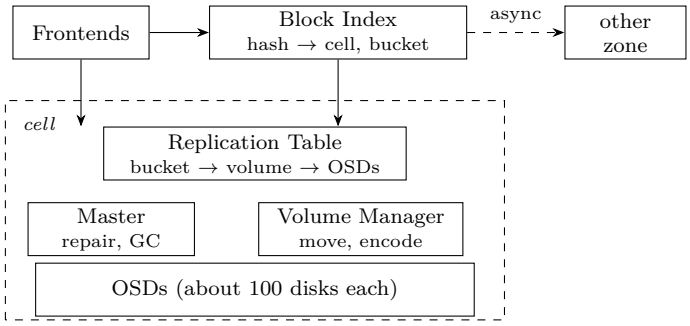


Figure 1: Magic Pocket within one zone, simplified from Cowling.[2] A cell holds roughly 50–100 PB; one Master per cell is what limits its size. Every block is copied to a second of the three US zones within about a second.

schedules repair and garbage collection but stays off the read and write path, and Volume Managers do the heavy copying and encoding.

New writes go to volumes replicated four ways. When a volume fills it is closed and erasure-coded with an LRC-(12,2,2) code: about 1.33× overhead against 1.5× for Reed–Solomon (6,3), while still surviving any three failures in a group.[10,12]

Durability as a process

Cowling's "Pocket Watch" post is the least cited and most useful source here.[3] A Markov model of disk failure and repair gives 27 nines, but models leave out software bugs and operator mistakes, so Dropbox runs six independent verifiers, and more than half of all disk and database load is this self-checking. At today's scale there are hundreds of hardware failures a day and about four 1–2 GB extents are repaired every second, within a 48-hour SLA.[10]

The six verifiers:[3] a Storage Watcher re-reads 1% of new blocks after a minute, hour, day, week and month; a Metadata Scanner checks about a million blocks a second; a Disk Scrubber reads every disk each one to two weeks; three more checkers make sure nothing is deleted without a log entry. Durability claims differ by source: 27 nines in the model, 12 on the blog,[1] 11 in the S-1.[8]

Lowering the cost per byte, 2016–2026

The design has kept changing. In 2019 a cold tier split old blocks in two, stored the halves in two regions and their XOR in a third. That cut the replication factor from 2× to 1.5×, saving 25% of disk, and tail latency actually improved because reads go to all three regions and take the first two answers.[4] Dropbox had first tried a single erasure code across regions and dropped it after nine months. From 2018 it moved to shingled (SMR) drives written as raw zones with no filesystem, about 20% more capacity per drive for roughly the same price;[5] by 2025 over 99% of the fleet was SMR, on drives of up to 32 TB.[6] Between the fourth and sixth server generations, each exabyte needed five fewer megawatts and a third of the rack space.[5] The newest change, in April 2026, fixed fragmentation left by a new on-the-fly erasure coder, with some volumes under 5% full, using a packing-based compaction step.[7]

What it saved, and what it did not

The S-1 is the only audited source.[8] It credits "Infrastructure Optimization" with cutting cost of revenue by $39.5M in 2016 ($92.5M

$M	2015	2016	2017
Revenue	604	845	1,107
Cost of rev.	407	391	369
Gross margin	33%	54%	67%
Capex	79	115	25

Table 2: From Dropbox's S-1.[8] Revenue nearly doubled while cost of revenue fell.

less paid to the "third-party datacenter service provider", offset by $53.0M of depreciation and facility costs) and by a further $35.1M in 2017. The often-quoted $74.6M is the sum of the two; the filing never states it. The same filing lists co-location sites in California, Texas and Virginia.

Three caveats keep the story honest. First, Dropbox never left AWS: the remaining data and some services stayed there, and from 2016 European business files have been stored on AWS in Germany.[1,13] Second, savings in cost of revenue leave out capital spending, about $219M over 2015–2017, and capital leases; Quinn uses exactly this to dispute a16z's claim that owned infrastructure costs a third to a half of cloud.[14,15] The FY2024 10-K still shows $92.7M a year of finance-lease cost for infrastructure.[9] Third, the case depends on one large, predictable, storage-heavy workload, which most companies do not have.

Plan for Weeks 2 and 3

We follow the cost and repatriation exercise. We will price a 1 EB, mostly cold blob workload on S3 and one other provider using their calculators, then build a bottom-up on-premises estimate (SMR drives, servers, power at about 0.3–0.5 W per TB,[5] co-location and staff). On a small test cluster we will also measure the storage overhead of three-way replication, erasure coding and the 2+1 XOR scheme.

Sources found

Tags: P primary (Dropbox or its filings), C talk or interview by Dropbox engineers, A academic, S secondary or commentary, V vendor.

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Corrections to our first draft. SMR was first deployed in 2018, not 2020; “500 million” counts sign-ups, not active users; $74.6M is a two-year sum we computed, not a figure printed in the S-1, and not an annual saving.